Marking Scheme:

Software CAT 3

Total: 50 marks

Accept any correct answer worded differently from the ones below.

1. API (2 marks)

- Definition (1): a set of rules/protocols that allows different software applications to communicate and exchange data.
- Example (1): any valid one, e.g. Google Maps API in a ride-hailing app, M-Pesa Daraja API for payments, a weather app pulling data from a weather service, "Login with Google".

2. REST vs SOAP (3 marks, 1 mark per valid difference)

- REST is an architectural style, while SOAP is a strict protocol.
- REST uses JSON, XML, HTML or plain text, while SOAP uses XML only.
- REST is lightweight and faster, while SOAP is heavier and slower.
- REST uses HTTP methods (GET, POST, etc.), while SOAP uses a single POST with an XML envelope.
- REST is stateless and cacheable, while SOAP has built-in security (WS-Security) and ACID compliance.
- REST is simpler to learn and use, while SOAP is more complex (WSDL contracts).

3. API endpoint (2 marks)

- Definition (1): a specific URL/address where an API receives requests and sends back responses for a resource.
- Example (1): e.g. https://jsonplaceholder.typicode.com/users or /api/students.

4. Features of Postman (4 marks, 1 mark each for any four)

- Sending HTTP requests (GET, POST, PUT, DELETE, etc.)
- Collections to organise and save requests
- Environments and variables
- Automated testing and test scripts
- Response viewer (status, headers, body, time)
- Mock servers
- API documentation generation
- Collaboration and team workspaces
- Monitoring and request history

5. HTTP methods (4 marks, 1 mark each)

- (a) View student records: **GET**
- (b) Add a new student: **POST**
- (c) Edit a student's details: **PUT** (or **PATCH**)
- (d) Remove a student record: **DELETE**

6. HTTP status codes (4 marks, 1 mark each)

- (a) 201: Created. The request succeeded and a new resource was created.
- (b) 401: Unauthorized. Authentication is missing or invalid.
- (c) 404: Not Found. The requested resource or URL does not exist.
- (d) 500: Internal Server Error. The server failed to process the request.

7. GET request in Postman (4 marks, 1 mark per step in logical order, any four)

1. Open Postman and create a new request (click New / + tab).
2. Select **GET** from the HTTP method dropdown.
3. Enter the URL https://jsonplaceholder.typicode.com/users in the address bar.
4. Click **Send**.
5. View the response (JSON list of users, status **200 OK**) in the response panel.

8. Corrective measures (3 marks)

- Corrective measures (1): actions taken after a fault or incident has occurred to fix the problem and restore normal operation.
- Preventive (1): actions taken beforehand to stop faults or incidents from happening, e.g. updates, backups, firewalls.
- Detective (1): actions that identify or discover that a fault or incident has occurred, e.g. monitoring, logs, intrusion detection.

9. Software corrective tools (4 marks: 0.5 for the tool and 0.5 for its purpose, per tool)

- Antivirus/anti-malware: detects and removes malicious software.
- System restore/recovery tools: return the system to an earlier working state.
- Backup and recovery software: restores lost or corrupted data.
- Disk repair utilities (CHKDSK, fsck): find and fix disk errors and bad sectors.
- Debuggers: locate and correct errors in code.
- Patch/update tools: fix known bugs and vulnerabilities.
- Data recovery tools: retrieve deleted or corrupted files.
- Uninstall/reinstall tools: remove faulty software.

10. Corrective steps on a faulty system (5 marks, 1 mark each for any five in logical order)

1. Identify and report the fault/symptoms.
2. Diagnose and isolate the cause (hardware or software).
3. Back up important data (if possible).
4. Plan and select the appropriate corrective action.

5. Apply the fix (repair, replace, reinstall, restore, patch).
6. Test the system to confirm the fault is resolved.
7. Document the fault and the action taken.

11. Software testing (3 marks)

- Definition (1): the process of evaluating software to find defects and check it meets requirements.
- Objectives (2, 1 mark each for any two): detect bugs/defects, verify the software meets requirements, ensure quality and reliability, validate that it works as expected, reduce the risk and cost of failure, ensure user satisfaction.

12. Smoke vs regression testing (2 marks, 1 mark each)

- Smoke testing: a quick, shallow check of the critical functions of a new build to confirm it is stable enough for further testing.
- Regression testing: re-testing existing features after changes, to confirm that fixes or new features have not broken them.

13. Levels of software testing (4 marks: 0.5 for the name and 0.5 for the description, per level)

- Unit testing: tests individual components or modules in isolation.
- Integration testing: tests how combined modules interact.
- System testing: tests the complete integrated system against requirements.
- Acceptance testing (UAT): the user or client confirms the system meets their needs before release.

14. Types of software maintenance (4 marks: 0.5 for the type and 0.5 for the example, per type)

- Corrective: fixing bugs, e.g. repairing a crash on login.
- Adaptive: adjusting to environment changes, e.g. updating the software for a new OS version.
- Perfective: improving performance or adding features, e.g. speeding up report generation.
- Preventive: preventing future problems, e.g. restructuring code or updating libraries.

15. Maintenance schedule (2 marks)

- Definition (1): a planned timetable of when maintenance activities (updates, backups, checks) will be carried out.
- Benefits (1, 0.5 each for any two): reduces unexpected downtime, extends system life, improves performance and reliability, reduces costs, helps with resource planning, improves security.